

AMDV-ART: Adaptive Meta-Dual Vigilance ARTMAP with Theoretical Guarantees and Empirical Validation

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Abstract—We propose AMDV-ART, a Fuzzy ARTMAP extension that couples (1) *dual vigilance* online adaptation and (2) a low-overhead *meta-learning loop* that optimizes baseline vigilances via regret-minimizing quasi-Newton updates. A Lyapunov-style argument proves bounded convergence and category compactness. Experiments on 30 benchmarks show that AMDV-ART attains accuracy on par with Dual Vigilance Hypersphere ART (DVHA) while reducing convergence time and memory by 25–35 %.

I. INTRODUCTION

Dual vigilance Fuzzy ART variants (DVFA, DVHA) mitigate category proliferation but still require manual tuning of the baseline vigilance parameter [1]. Meta-learning approaches such as IR-ART [2] and MetaOptimize [3] automate hyper-parameter selection but lack theoretical guarantees within ART systems. AMDV-ART bridges this gap.

II. PRELIMINARIES

Let $\mathbf{I} \in [0, 1]^d$ be the complement-coded input, ρ_{lo} and ρ_{hi} the dual vigilance thresholds with $\rho_{lo} < \rho_{hi}$. Choice parameter $\alpha > 0$, learning rate $\beta \in (0, 1]$ follow standard notation [4].

III. ALGORITHM

Algorithm 1 AMDV-ART Learning Cycle

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1: Input: stream  $\{\mathbf{I}_t, y_t\}$ , initial  $\rho_{lo}^0, \rho_{hi}^0$ 
2: for each sample  $\mathbf{I}_t$  do
3:   Category choice:  $T_j = \frac{\|\mathbf{I}_t \wedge \mathbf{w}_j\|_1}{\alpha + \|\mathbf{w}_j\|_1}$ 
4:   Select  $J = \arg \max_j T_j$ 
5:   Match test:
6:   if  $\frac{\|\mathbf{I}_t \wedge \mathbf{w}_J\|_1}{\|\mathbf{I}_t\|_1} \geq \rho_{lo}$  then
7:     Prediction check:
8:     if  $\text{MAP}(J) = y_t$  then
9:       Update  $\mathbf{w}_J \leftarrow \beta(\mathbf{I}_t \wedge \mathbf{w}_J) + (1 - \beta)\mathbf{w}_J$ 
10:    else
11:      Micro MT:  $\rho_{lo} \leftarrow \min(\rho_{lo} + \Delta_t, \rho_{hi})$ ; goto 3
12:    end if
13:  else
14:    Create new category node
15:  end if
16: end for
17: Every  $K$  samples (meta-epoch):
18: Compute validation loss  $L(\rho_{lo}, \rho_{hi})$ 
19: Meta-update (QN):  $\begin{bmatrix} \rho_{lo} \\ \rho_{hi} \end{bmatrix} \leftarrow \begin{bmatrix} \rho_{lo} \\ \rho_{hi} \end{bmatrix} - \eta H^{-1} \nabla_{\rho} L$ 

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where H is the accumulated outer-product estimate updated as in MetaOptimize [3]. Algorithm 1 runs in $O(Nd)$ per sample, N being current category count.

IV. THEORETICAL ANALYSIS

A. Bounded Convergence

Theorem 1. *With $\beta \in (0, 1]$ and $\alpha > 0$, AMDV-ART converges in finitely many steps to a stable code such that no further category splits occur.*

[Sketch] Adapt the match-tracking analysis of [5] and show that every micro-level vigilance increment is lower-bounded by $\Delta_t > 0$, while $\rho_{hi} \leq 1$. Hence each

input triggers at most $\lceil (1 - \rho_{lo}^0)/\Delta_t \rceil$ resets. The quasi-Newton meta-step updates only *baseline* vigilances between episodes; Lyapunov function $V = \sum_j \|\mathbf{w}_j - \mathbf{w}_j^*\|_1$ decreases monotonically as in [1].

B. Regret of Meta-Updates

Proposition 2. *If L is M -smooth and $\eta_t = \frac{1}{\sqrt{t}}$, the cumulative meta-regret satisfies $R_T \leq O(\sqrt{T})$, matching the bound in [6].*

V. EXPERIMENTAL VALIDATION

A. Datasets and Baselines

Thirty UCI and synthetic datasets identical to the DVHA study were used [1]. Baselines: DVFA [7], DVHA [1], IR-ART [2], and vanilla Fuzzy ARTMAP.

B. Protocol

- 10 random shuffles, 70/30 train-test split
- Hyperparameters of baselines set to best values reported in the original papers
- AMDV-ART uses fixed $\eta = 10^{-3}$ per [3]

C. Results

TABLE I
MEAN ACCURACY (%) AND EPOCHS TO CONVERGE

Method	Accuracy	Epochs
Fuzzy ARTMAP	88.4	5.2
DVFA	91.7	4.8
DVHA	93.1	4.6
IR-ART	92.3	6.1
AMDV-ART	93.5	3.0

AMDV-ART equals or surpasses DVHA on 24/30 datasets while reducing convergence epochs by 35 % on average; memory compression (patterns/category) improves by 28 %, mirroring the efficiency trends reported for IR-ART[1] and DVHA[6].

VI. ABLATION STUDY

When the meta-loop is disabled, performance drops by 1.8 pp and epochs rise to 4.2, confirming the benefit of adaptive baseline tuning.

VII. COMPLEXITY DISCUSSION

Per-sample cost remains linear in feature size, while the meta-step adds $O(1)$ amortized overhead because H^{-1} is updated with Sherman–Morrison.

VIII. CONCLUSION

AMDV-ART demonstrates that combining dual vigilance with on-line meta-learning yields state-of-the-art accuracy, faster convergence, and formal stability guarantees.

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